

HALCYON CREDIT

Agentic Underwriting Copilot

Production Readiness Review and Submission Report

Futureense x IIT Gandhinagar Project 02 Week 4.5 of 6

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Every figure in this report is reproducible from committed code: `python -m eval.golden_set | eval.benchmark | eval.fairness | eval.economics`

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1. Executive summary

We are building the agentic reasoning, verification, policy and evaluation layer of a credit underwriting copilot. We are not building the risk model, which is a commodity tool that sits inside it. In the last two sprints the project moved from docs only to a runnable, safety benchmarked, end to end system on a real 1.6 million row dataset, with an evaluation harness that produces the project's headline result.

Honest completion, grade weighted: about 60 percent. The two heaviest and previously empty buckets, Engineering (25 percent) and Quality and Evaluation (25 percent), went from near zero to majority done. What remains is deployment, the live LLM wiring, and the final evaluation polish (RAGAS, judge calibration, red team), all scheduled into the 1.5 weeks left. We report real numbers rather than a round headline.

Headline result. On a 128 case labelled golden set, the agentic pipeline catches 100 percent of regulatory hard stops and commits zero dangerous over approvals, versus 26.5 percent and 16 for a single LLM baseline. It also grounds every decision with citations the baseline cannot produce.

Snapshot

Metric	Value
Weighted readiness (honest)	about 60 percent (Engineering and Quality/Eval up from about 16 percent)
Dangerous over approvals (agentic vs baseline)	0 vs 16
Regulatory hard stop catch (agentic vs baseline)	100 percent vs 26.5 percent
Real merged training table	1,613,898 rows, 0 percent feature nulls
Automated tests green this session	16 (safety invariants CI gated)

2. What we are building

One raw application flows through a typed, checkpointed graph. Three worker agents (income, credit, policy) run in parallel; a barrier merges them; the risk model scores; a synthesizer drafts a grounded recommendation; an evaluator gates it on faithfulness with a capped retry loop; and a record writer persists a replayable audit trail.

Design thesis: the only irreversible, large blast action, disbursing a loan, sits outside the agent, behind a human gate. The copilot advises; the underwriter decides. Agency is scoped to reasoning, never to irreversible action.

3. Sprint reviews

Sprint 1: Design and De-risk (status: about 95 percent complete)

Delivered	Evidence
System (HLD), sequence (LLD) and state schema diagrams	docs/diagrams/, Agentic Architecture.png
Typed, schema first shared state with agent write isolation	state/application_state.py
8 agent contracts plus failure mode analysis	agents/nodes.py
Dataset defended vs 8 alternatives; North Star = Trusted Decision Rate	docs/DATASET_COMPARISON.md, ADR-005/006
12 Architecture Decision Records	DECISIONS.md (ADR-001 to 012)
Repo scaffolding, CI stub, risk register, runbook	.github/workflows/ci.yml, docs/

Carried forward: LiteLLM gateway to live models; policy corpus RAG spike.

Sprint 2: Build Core (status: about 70 percent complete, was 0 percent)

Delivered	Evidence
Runnable vertical slice (parallel, merge, risk, synth, eval loop, record)	graph.py, run_slice.py
Audit replay: about 10.7 recoverable checkpoints per decision	MemorySaver in graph.py
Real merged dataset: 1,613,898 rows (HC 8.07 percent, LC 20.09 percent; 0 percent nulls)	data_pipeline/, data/merged_dataset_card.md
Leakage guard plus OOM safe chunked reader for the 1.2 GB file	lendingclub_adapter.py, ADR-012
Golden set: 128 cases, 8 strata, independent oracle	eval/golden_set.py, eval/oracle.py
Single LLM baseline plus agentic vs baseline benchmark	eval/baseline.py, eval/benchmark.py
Fairness audit on real data (gender and age)	eval/fairness.py
Cost and economics model plus Production Readiness Scorecard	eval/economics.py, docs/READINESS_CARDS.md
Safety invariants gated in tests (16 passing this session)	tests/test_eval.py, tests/test_merge.py

4. Progress against the rubric

Bucket	Weight	Start	Now	To reach target
Architecture	20%	~80%	~85%	failure mode drill in demo
Product	15%	~75%	~80%	tie cost and value story to PRD KPIs
Engineering	25%	~28%	~55%	LiteLLM live, API and UI, deploy, CI gate

Bucket	Weight	Start	Now	To reach target
Quality and Eval	25%	~5%	~55%	RAGAS, calibrated judge, red team, DSPy
Demo	15%	0%	~20%	deployed URL, 20 minute video
Individual viva	gate	at risk	on track	each owner rehearses their vertical

Weighted readiness is about 60 percent. Path to 80 percent (final 1.5 weeks): LiteLLM live plus real baseline numbers; policy corpus RAG spike; deploy and record demo; RAGAS and judge calibration; production risk model and calibration curve.

5. Production Readiness Scorecard

Card 1: Agency

The control flow has 12 decision edges, classified by who decides: R for rule, H for human, M for model.

Field	Value
Total edges	12
R edges (rule decides)	5: orchestration fan out; policy hard stop; merge barrier; retry router; escalate trigger
H edges (human decides)	1: the binding approve or decline (copilot design; agent output is advisory)
M edges (model decides)	6: income read; credit interpretation; policy RAG; risk band; decision synthesis; faithfulness judge
Agency present?	Yes
Scale of agency (M share)	50 percent, Medium (0 binding or irreversible edges)
Leash (longest unsupervised run)	3 boxes: risk, synthesize, evaluate (all checkpointed)
Riskiest M edge	the synthesize decision, i.e. the approve, decline or refer draft

Verdict: yes, this problem deserves agency. Reconciling heterogeneous thin file evidence and grounding it against policy is genuinely multi step. But every model edge is bounded by typed state, mandatory citations, a faithfulness gate, deterministic hard stops and a human on the binding call, so agency is scoped to reasoning, never to irreversible action.

Card 2: Before and After (blast radius by reversibility)

The agent's model edges by risk quadrant: Reversible and Small = 6; Reversible and Large = 0; Irreversible and Small = 0; Irreversible and Large = 0. The one irreversible and large action, loan disbursement, is human gated and sits outside the agent.

M edge	Worst case	Quadrant	What we built	Gap?
synthesize decision	Confidently approves a should decline; underwriter rubber stamps	Rev and small (consequential)	Citations required, faithfulness gate, human sign off, so 0 over approvals / 128	Judge calibration (Ph 3.2)
policy hard stop	Misses underage, over LTI or prohibited clause	Irrev and large if acted on	Deterministic rule, not the model, so 100 percent catch (0/49 missed)	Real policy RAG (Ph 2.1)
risk band	Mis scores, so wrong band	Rev and small	Fairness measured on real data (DP 6.0pp; age 30 to 74 percent)	Production GBM (Ph 2.3)
income or credit read	Wrong verified figure	Rev and small	Verified (not stated) income plus discrepancy flag	Real data sources
escalate to human	Fails to route low confidence out	Rev and small	escalated=True when faithfulness below 0.80 after retries	None

Cards 3 and 4: Economics (unit is one loan decision)

Prices: GPT-4o-mini and GPT-4o list rates (July 2026), FX 83 rupees per USD.

Input	Happy	Average	Sad
Outcomes per month	10,000	30,000	60,000
Model calls per outcome	5	5.4	9
Retries per outcome	0	0.2	2 (max)
H gate trigger by minutes	8% x 3	15% x 4	40% x 6
Cost per outcome	Rs 4.91	Rs 10.63	Rs 40.82
Total monthly cost	Rs 49,064	Rs 3,18,865	Rs 24,49,089

Read: token cost is trivial (1 to 4 rupees per decision); the human review gate dominates the bill. The cost lever is the escalation rate, which is exactly what the faithfulness gate controls.

Card 5: Business value

	Happy	Average	Sad
Total monthly cost	Rs 49,064	Rs 3,18,865	Rs 24,49,089
Faithfulness >= 0.80 (100 percent pass)	Y	Y	Y
Hard stop catch = 100 percent	Y	Y	Y
0 dangerous over approvals	Y	Y	Y
Decision accuracy >= 75 percent (78.1)	Y	Y	Y
Fairness gap within threshold	measured	watch	watch
SLAs held (latency, availability)	Y	Y	N

Verdict: the Average configuration wins. With cheap and strong LiteLLM routing, retry cap 2, faithfulness gate 0.80 and about a 15 percent human gate, it meets all quality KPIs and SLAs at 10.63 rupees per decision (about 3.2 lakh rupees per month at 30k decisions). The Sad path shows where to add an async human review queue plus fallback budget before scaling past about 60k per month.

6. Headline benchmark: agentic vs single-LLM

A 128 case golden set (8 strata, seed 42), graded against an independent senior underwriter oracle. Both decision engines are deterministic stand ins; this isolates the structural safety the agentic design adds, before real LLM wiring. Integrity note: model IQ deltas require the real LiteLLM baseline (Phase 1.5).

Metric	Agentic	Single-LLM	Why it matters
Decision accuracy vs oracle	78.1%	55.5%	overall correctness
Hard stop catch rate	100%	26.5%	regulatory declines never missed
Hard stop violations (of 49)	0	36	each is a compliance breach
Dangerous over approvals	0	16	approving a should decline or refer
Grounding (source_refs per decision)	1.12	0	every claim cited
Mean faithfulness (pass rate)	1.0 (100%)	n/a	cited claim ratio ≥ 0.80
Audit checkpoints per decision	10.7	0	regulator replayable

The accuracy edge (plus 22 points) is modest and honest; the safety edge is decisive. On ambiguous middle strata the agent can trail a conservative human, which is a calibration gap (Phase 3.2), not a reason for more agency.

7. Fairness audit on real data

300,000 real Home Credit applicants. A fairness through unawareness probe (protected attributes excluded from the features) still inherits base rate gaps, so we measure and surface them per decision rather than assume unawareness equals fairness.

Measure	Value
Default rate by gender	F 7.0 percent, M 10.1 percent
Default rate by age	under 25: 12.4 percent; 25 to 34: 10.7 percent; 35 to 49: 8.1 percent; 50 plus: 5.7 percent
Approval rate by gender	F 67.3 percent, M 61.3 percent
Demographic parity gap (gender)	6.0 pp
Equal opportunity gap (gender)	4.9 pp
Approval rate by age	under 25: 29.6 percent up to 50 plus: 73.6 percent (strong gradient)

These gaps are real and not yet mitigated; the Phase 3 fairness gate (reweighing or per group thresholds) is the remediation. Measuring and surfacing them is itself the deliverable.

8. Contributions

Collective (this fortnight)

- Took Engineering and Quality/Eval from about 16 percent docs only to a runnable, tested, benchmarked system on real data, the two 25 percent buckets that decide the grade.
- Produced the headline evaluation result (agentic vs single LLM) with honest, reproducible deltas.
- Ran a real fairness audit on 300k applicants and surfaced concrete gender and age gaps.
- Built a bottom up cost model: 4.91 to 40.82 rupees per decision, with the human gate (not tokens) as the lever.
- Completed the Production Readiness Scorecard with numbers traceable to committed code.

Individual ownership (each member defends their own vertical at the viva)

Ayush Tiwary: Evaluation, Policy grounding and Architecture spine

Golden set and oracle, agentic vs baseline benchmark, faithfulness gate, fairness audit; ADR log and Master Build Plan; Readiness Scorecard.

Viva: how do we know the agent is trustworthy, and how much better than one prompt?

Himkar Vashishta: Model Gateway, Synthesis and Economics

LiteLLM cheap and strong routing, Decision Synthesizer contract, cost and economics model, single LLM baseline; HLD, LLD and schema diagrams.

Viva: what does each decision cost, and how is the model call bounded?

Aditya Arora: Orchestration, State and Persistence

LangGraph graph (fan out, barrier, retry loop), typed ApplicationState, DecisionRecord and audit replay, CI, forthcoming FastAPI.

Viva: how does the graph run, recover, and produce a replayable audit trail?

Harshit Gautam: Data and Risk Model

Dataset selection and EDA, HC and LC merge pipeline (canonical schema, adapters, leakage guard) to a 1.6M row table; risk kernel; PRD, TRD and research.

Viva: why this data, is it leakage free, and how is risk scored and explained?

9. Path to a top-grade submission (final 1.5 weeks)

1. LiteLLM live plus real single LLM baseline numbers.
2. Policy corpus ChromaDB RAG spike (grounded retrieval with source_refs).
3. Deploy to a free tier plus record the 20 minute demo.

4. RAGAS runner plus LLM judge calibrated against human labels.
5. Production risk model: GBM, SHAP and calibration curve.
6. Fairness mitigation gate plus CI regression gate on the golden set.

10. Evidence index (all committed, all reproducible)

Claim	Source
Edge counts, agency scope	graph.py, agents/nodes.py
Benchmark (78.1 vs 55.5 percent; 0 vs 16 over approvals)	eval/benchmark.py, eval/benchmark_report.md
Golden set (128 cases, 8 strata)	eval/golden_set.py, eval/oracle.py
Fairness (gender DP 6.0pp; age 30 to 74 percent)	eval/fairness.py, eval/fairness_report.md
Economics (4.91 / 10.63 / 40.82 rupees per outcome)	eval/economics.py
Safety invariants gated in CI	tests/test_eval.py (6 tests)